

Session 221: Communication Signal Processing and Coding

TIME 2:30 PM – 4:00 PM IST VENUE SP 1.20 (TVS Lab)



221.1	Hardware-Efficient FPGA Implementation of OTSM Transceiver for High-Mobility 6G Communications Vishnupriya P V, Sapta Girish Neelam
221.2	Iteratively reweighted algorithms for robust sparse recovery with l_p ($0 < p < 1$)-norm minimization Shweta Bharti, Samrat Mukhopadhyay, Govind Murmu
221.3	Coded SumComp: Linear Coding over Integer Rings for Improved SumComp Performance Vaishnavi Anekar, Anjana A M
221.4	A Novel Tunable Window for Signal Processing Applications Debanjan Bhattacharjee, Sujit Kumar Sahoo
221.5	PYNQ-Based Modular Architecture for Real-Time Stream-Cipher Video Encryption at the Edge Vanukuru Hari Rohit, D. Sreekiran, Raghavendra Kune, D. Mallikharjuna Rao, Novaline Jacob
221.6	GSM-Based Non-Orthogonal Multiple Access: Design and Evaluation Surjeet Kumar, Ashish Kant Shukla, Manav R Bhatnagar
221.7	Enhancing Power Efficiency in MIMO DFT-s-OFDM: A Relative Study of $\pi/2$-BPSK with Tone Reservation Abhinaba Dey, Sreenath Ramanath
221.8	Molecular Communication with Finite-Length Transmitter in Time Dependent Diffusive Channel Neha Bhusari, Avinash G. Keskar, Nikhil Sharma, Prabhat Kumar Sharma
221.9	Generalized Partial Full-Pilot Zero-Forcing Combining for Distributed Massive MIMO Mohd Saif Ali Khan, Karthik RM, Samar Agnihotri
221.10	Signal Space Decoupled Stream Splitting for Low Latency Downlink MIMO WBANs Gunjit Arora, Satya Kumar Vankayala, Manoj Thomas
221.11	Low Complexity Digital Predistortion for Power Amplifiers Using Incremental PCA Girish Tripathi, Anindya Saha
221.12	Rank-Deficit Hybrid Adaptive Fulcrum Decoder for Low-Complexity Network Coding Sovanjyoti Giri, Satya Prakash
221.13	Evolutionary Optimization for Quantized RIS Phase Shift Design in MIMO Systems: Hyperparameter Analysis and SNR Performance Amit Shrivastava, Manish Kumar, Kunwar Pritiraj Rajput
221.14	Deep Unfolded AMP for Short Block-Length SPARC over AWGN Channels Sai Dinesh Kancharana, Sandeep VSV, Arun Pachai Kannu
221.15	Detection in Ambient Backscatter Communication System with Multi-Antenna Tag-Reader Pair with Correlated Ambient Source Signal Anuranjan Jha, Adarsh Patel
221.16	Regularized Multi-Measurement Sparse Decomposition for Time-Domain Ambisonics Upscaling Gyanajyoti Routray, Priyadarshini Dwived, Rajesh M. Hegde

Session 222: Vision and Sensor Analytics

TIME 2:30 PM – 4:00 PM IST VENUE SP Building Side Foyer



- 222.1

Signal-driven Shipwreck Segmentation and Probabilistic Uncertainty Estimation with Sidescan Sonar for Autonomous Underwater Navigation
Lakshmi Annamalai
- 222.2

Vision-Powered Cleaning Mode Detection in Robot Vacuum Cleaners Using a Novel Smart Home Dataset
Sanket Gharatkar, Suhel Khan, Nagaraj Yendeti
- 222.3

NuSegFormer: Enhanced Cell Nuclei Segmentation Using a Hybrid UNet--Vision Transformer with Gated Multi-Head Attention
Aditi Damodar, Hitesh Tekchandani, Praveen Kumar Shukla, Anish Kumar Vishwakarma
- 222.4

On the Optimal Number of Grids for Differentially Private Non-Interactive K-Means Clustering
Gokularam Muthukrishnan, Anshoo Tandon
- 222.5

Are Wavelets More Reliable? Comparing The Topology of Image Generator Metrics
Manas Patil, Amit Tiwari, Abhijat Bharadwaj, Vikram Gadre
- 222.6

Domain-Adaptive Self-Supervised Pretraining for Bacterial Flagellar Motor Detection in Cryo-Electron Tomography
Adavi Jayanth, Madhu Oruganti
- 222.7

Enhancing Road Safety through Real-Time Unsupervised Anomaly Detection in Crash Barriers
Divya Varshney, Avnish Panwar, Indrajit Ghosh
- 222.8

Multi-Sensor Alignment for Weather Simulations
Samsad Alam, Devyani Lambhate, Aditya Mohan, Vishal Kumar, Vaibhav Katewa
- 222.9

SIADS: Innovation-Based Anomaly Detection in Streaming Communication Graphs
Tanish Verma
- 222.10

LS-PolypSeg: A Parameter-Efficient LoRA-Enhanced SAM3 Framework for Polyp Segmentation
Sanjana Jhansi Ganji, Srikanth Panigrahi, Kaushal Sambanna, Srinivas Rao Routhu
- 222.11

Tube Loss-based Detection of Anomalies in Electricity Consumption Time Series Data
Shruti Mehta, Pritam Anand, Yash Vasavada
- 222.12

Learning What to Trust: Confidence-Aware Curriculum Distillation for TinyML Vision
Satarupa Das, Soumen Garai, Rajrup Saha, Soumya Chatterjee, Suman Samui

Session 223: Wireless Resource Optimization

TIME 2:30 PM – 4:00 PM IST VENUE ECE 1.07



223.1	Mobile Base Station Optimal Tour in Wide Area IoT Sensor Networks Sachin Kadam
223.2	Robust Deep Reinforcement Learning for Task Scheduling in MEC-Enabled IoT Networks Vikky Masih, Arghyadip Roy
223.3	DPAS: Demand-Pressure Adaptive Slicing for Real-Time 6G Network Resource Management Jaynil Panjwani, Neha Agrawal
223.4	Ergodic Throughput in Coordinated Multi-point Downlink under Distributed Power Constraints mahesh nagulapalli, Mahendra Pratap Singh, Anil Kumar, Sibi Raj B. Pillai
223.5	Mobility-Aware Offloading with SINR Prediction and Selective Replication in Fog-IoT Networks Rajasekhar Dasari, Satyajit Mohapatra, Sanjeet Kumar Nayak
223.6	Decoding-Order-Aware Secrecy Rate Maximization in IRS-Assisted Untrusted NOMA Under Imperfect SIC Ankit Kumar, Kaneez Khatoon, Deepak Mishra, Ravikant Saini
223.7	Reinforcement Learning-Based Decode and Forward UAV Relay Trajectory Optimization aniket Subbanwar, Ojas Joshi, Amit Agarwal
223.8	Measurement-Based Energy-Efficiency Optimization for On-demand Data Rate Provisioning in 5G RAN Ramagiddaiah Eediga, Preethi G V, Anand S V R, Chandra R. Murthy, Ryan Husbands
223.9	Efficient Sampling for Pure Exploration Problems Dhinesh Chandran G, Srinivas Reddy Kota, Srikrishna Bhashyam
223.10	Link Adaptation Using Joint-Thompson Sampling Vignatha Vinjam, Manjunath Kolavennu, Myna Vajha, Karthik Periyapattana Narayanaprasad
223.11	Adaptive Scheduling for Low-Latency Coordination in Wi-Fi 8 Multi-AP Networks Hiya Shah
223.12	Robust Base Station Placement in Agricultural IoT via Bayesian Optimization Gourav Prateek Sharma, Durgesh Singh, James Gross